SnapSafe AI — Private

“See it. Detect it. Protect it. Before you sh

- Platform Target: HP Snapdragon AI PCs
- Track: On-Device AI, Privacy & Security,
- Core Differentiator: 100% On-Device Infe

AI Protection for Your Screen

are it.”

(Snapdragon X Elite / Snapdragon X Plus)

Real-Time Utility

rence • Exactly 0 Bytes Cloud Egress

The Problem: Screen Shari

The invisible privacy crisis during remote prese

- Developers accidentally stream IDE tabs contai
- Students expose personal student IDs, phone n
- Cloud AI Fallacy: Existing assistants stream full
- x86 CPU Penalty: Continuous vision models co

ng Data Leaks

ntations

ning live API keys, AWS credentials, and .env tokens.

umbers, and academic records during classroom shares.

screenshots to cloud servers, introducing massive privacy risks.

nsume 35W+ on legacy PCs, draining battery and spinning fans.

The Solution: SnapSafe AI

An intelligent on-device privacy firewall before screen

- Local Screen Capture: Acquires display buffers in-mem
- Neural Text Extraction: High-precision tokenization and
- Multi-Vector Detection: Identifies API keys, passwords,
- Smart Redaction: Irreversible Gaussian blur, pixelation,
- Safe Share: Before/After split comparison slider demon

broadcast

ory with sub-20ms latency.

spatial bounding boxes.

student/employee IDs, phone numbers, and cards.

or solid blackout masking.

strating exposure plunge (Risk 87 → 4).

Product Workflow

5-stage on-device pipeline

- 1. Capture Screen Buffer: Uncompressed displ
- 2. Local OCR Model: Runs DBNet + CRNN via
- 3. Sensitive Classification: Shannon entropy (>
- 4. CVSS Risk Scoring: Assigns Critical / High /
- 5. Safe Export: Exports protected PNG or clipb

ay frame retained strictly in RAM.

ONNX / Qualcomm QNN execution providers.

3.2 bits/char) and Luhn Mod-10 card validation.

Medium / Low tiers; computes Privacy Score (0-100).

oard buffer for zero-risk screen broadcast.

Competition Live Demo

One-click pitch demonstration

- Synthetic Academic Portal: Generates realistic
- Instant Detection: Catches OpenAI key, passw
- On-Device Privacy Coach: Explains specific th
- Interactive Comparison: Judges drag slider to
- Guaranteed Synthetic Data: 0 real credentials

student record with synthetic test tokens.

ord, student ID, mobile number, and confidential note.

reat vectors and recommended mitigations.

reveal original vs protected redacted screen.

or real personal data used in demonstrations.

Technical Architecture

Modular, decoupled, and hardware-neutral

- Frontend: React 18, TypeScript, Vite, Tailwind CSS, L
- Backend: Python FastAPI local loopback engine (127.
- Desktop: Electron shell with native Windows frameles
- Audit Store: SQLite metadata-only log with strict zero-
- Verification: 25 automated unit tests with 100% pass r

ucide icons.

0.0.1:8000).

s integration and display capture IPC.

raw-pixel retention.

rate across detection, risk, and redaction.

Snapdragon Optimization & Q

Engineered for Qualcomm Hexagon NPU acceleratio

- Hexagon NPU Acceleration: Designed to offload vision
- Sub-5W Efficiency: Operates silently without fan noise
- Qualcomm AI Hub Abstraction: Pluggable BaseOCRP
- Execution Provider Hook: Binds directly to onnxruntim
- Truthful Disclosure: Prototype host runs local CPU fall

ualcomm AI Hub

on

n models to Snapdragon's 45 TOPS NPU.

e or thermal throttling during long meetings.

rovider ready for compiled QNN context binaries.

e-qnn with QNNExecutionProvider.

back; ready for immediate NPU deployment.

Privacy & Security Model

Zero-trust local processing guarantee

- Cloud Egress Invariant: Exactly 0 bytes upload
- No External AI Calls: Zero reliance on cloud LL
- Ephemeral RAM Processing: Raw unredacted
- Irreversible Redaction: Mathematical destructi
- Security Audit: Zero hardcoded credentials, ze

ed to external servers.

.Ms or third-party vision APIs.

display buffers are scrubbed immediately from memory.

on of glyph strokes prevents deconvolution recovery.

ro telemetry of sensitive strings.

Market Impact & Target Audi

Broad everyday utility across professional and aca

- Students & Universities: Prevents accidental roll nu
- Software Developers: Eliminates leaked production
- Technical Interviewees: Protects proprietary docum
- Telehealth & Remote Support: Enforces local comp

ence

demic sectors

mber and transcript leaks during online presentations.

API keys during live hackathon demos and reviews.

ents and personal contact info from stream leaks.

liance guardrails for healthcare and enterprise staff.

Roadmap & Conclusion

“SnapSafe AI turns the AI PC into a privacy

- Phase 1 (Current): Fully verified local MVP
- Phase 2 (Target): Direct Qualcomm AI Hub
- Phase 3 (Target): Windows Virtual Display
- Closing: Privacy is the killer use case for Sr

firewall.”

with OCR, multi-vector detection, redaction, and pitch demo.

QNN binary context compilation on Hexagon HTP v75.

Driver (WDDM) for continuous 60 FPS background protection.

apdragon AI PCs. See it. Detect it. Protect it. Before you share it.